PROJECT 1 - STATISTICAL LEARNING

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Introduction

This project focuses on analyzing diabetes disease using various data analysis techniques. The data used for it is called "diabetes.csv" which initially contains 9 variables and 768 observations. The variables are:

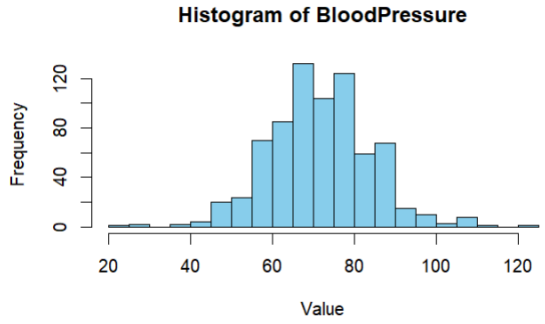
Pregnancies: number of pregnancies had.
Glucose: glucose level.
BloodPressure: blood pressure level.
SkinThickness: skin thickness.
Insulin: insulin level.
BMI: body mass index, derived from height and weight.
DiabetesPedigreeFunction: probability of having diabetes based on family history.
Age: Participant's age in years.
Outcome: 1-Participant has diabetes. 0-Participant doesn't have diabetes.

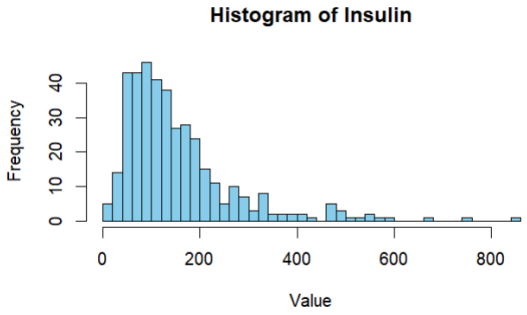
Data preprocessing

First, we changed the variable *DiabetesPedigreeFunction*'s name to *DPF*. Upon examining our data, we observed there were no NA or missing values, instead, those were represented as "0". To address this first we obtained the number of 0-values.

```
Glucose BloodPressure SkinThickness       Insulin           BMI
      5            35           227           374            11
```

To deal with missing values, we need to check the skeweness of the variables with NA. A variable is considered skewed when its distribution is not symmetric(the values are concentrated more on one side, resulting in a longer tail on either the left or right). It isn't skewed when the value if close to 0.



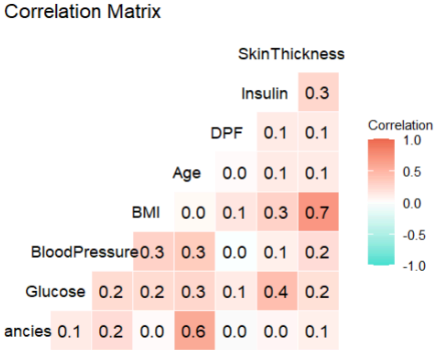


We plot an histogram for each variable and we can consider that BloodPressure(`0.1336042`) and Glucose (`0.5289026`) aren't skewed and in the other hand Insulin(`2.149996`), SkinThickness(`0.6867941`) and BMI(`0.5916179`) are skewed to the right. We use knn to deal with missing values in skewed variables and we just replace the missing values with the mean in the other variables.

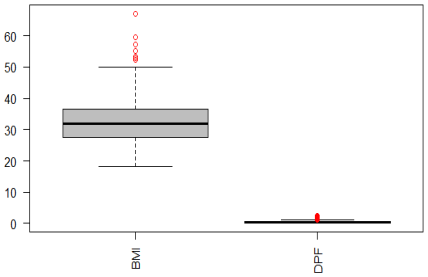
```
        0         1
0.2516129 0.7567568
```

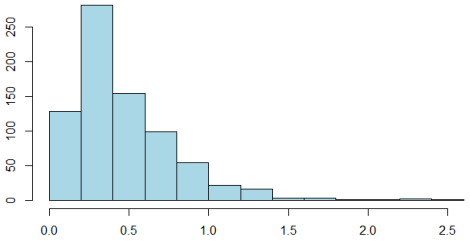
A new binary variable *HighRisk* was created, defined as 1 when *Glucose* > 140 and *BMI* > 30, and 0 otherwise, to capture the joint effect of obesity and high glucose levels. Using the function tapply we conclude that individuals classified as high-risk are roughly three times more likely to have diabetes.

We used a correlation matrix (only numeric variables) to visualize strong positive relations between the variables. The most notable ones were between *SkinThickeness* and *BMI* **(0.7)**, *Insuline* and *Glucose***(0.4)**, and *Pregnancies* and *Age* **(0.6).** This last one may not be so statistically relevant but it is still noticeable in the matrix.
(More data visualization in the notebook)



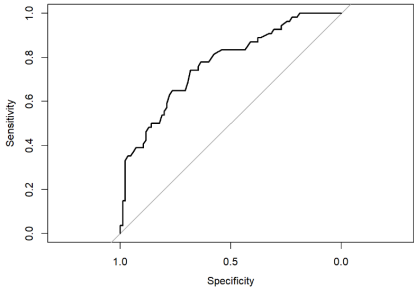
The next step was dealing with outliers. To get a general idea, we created a boxplot which visualizes the distribution of the continuous variables.
To come up with a method to identify the outliers we analyzed the distribution of *DPF*.





DPF is clearly skewed to the right (the distribution is not normal), therefore, the IQR rule is the better choice for identifying outliers in this dataset, against the Sigma rule which may not work well with skewed data, as it assumes symmetry around the mean. Once we identified outliers we created a new dataset "*clean_data"* where they were removed.

Linear and Logistic Regression & Resampling

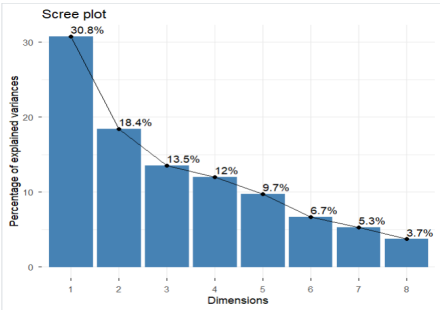



```
ROC        Sens       Spec
0.7990777  0.8923219  0.5021739
```

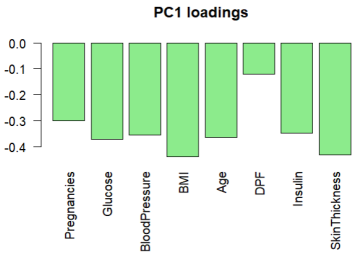
Logistic regression models were fitted using each predictor variable against the outcome (diabetes) to identify which variable best predicts the disease. Among all the predictors analyzed, glucose emerged as the strongest individual predictor, providing the best results in both discrimination ability and overall classification metrics. The model using glucose achieved a high area under the ROC curve (about 0.77–0.80), demonstrating its effectiveness at distinguishing diabetic from non-diabetic subjects. Glucose also yielded high sensitivity and acceptable specificity, indicating that it can detect most cases of diabetes while maintaining a reasonable balance in identifying non-cases. These results confirm that blood glucose level is a key factor for diabetes prediction, making it the most important variable among those tested (more detail in notebook).

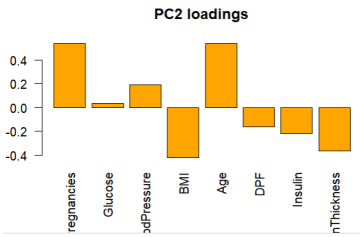
Principal Component Analysis (PCA)

We performed PCA on the clean dataset (without outliers). First we began by scaling the data, selecting our predictors and then computed a screenplot which shows the percentage of explained variances by each variable.



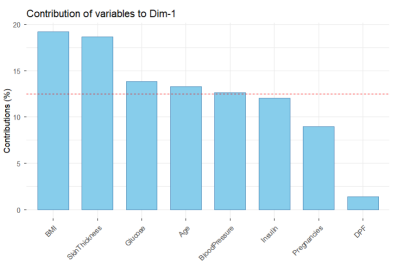
The graphs below help to visualize the loadings for PC1 and PC2, the variables with larger magnitude define the meaning of each PC.

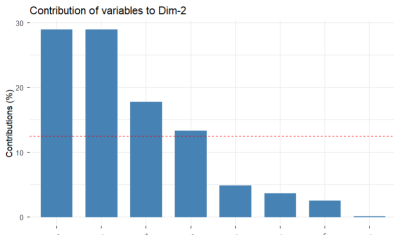




The following graphs represent which of the variables explain most of the variance of the data (those above the line).

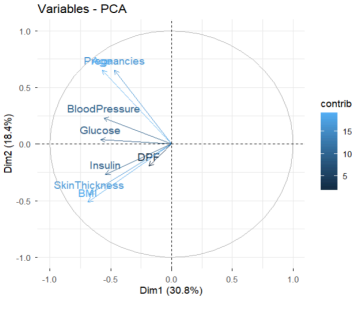
- 1st PC (left graph): *BMI* and *Skinthickness ,* closely followed by *Glucose.*
- 2nd PC (right graph): *Pregnancies* and *Age and BMI.*

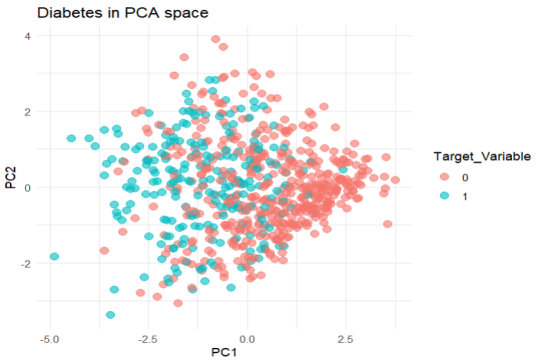




This means that these variables are the most influential in describing differences between patients, and should be prioritized when analyzing patterns or building predictive models for this data.

The graph to the right is used to show how much each variable contributes to the principal components. The darker the color, the higher the contribution, as is for variables *DPF, Insulin, Glucose* and *BloodPressure.* In the R script we also computed Biplots, which show PCs as axes, arrows for variables (directions of increasing values). Arrows close together indicate positively correlated variables; opposite arrows indicate negative correlation.

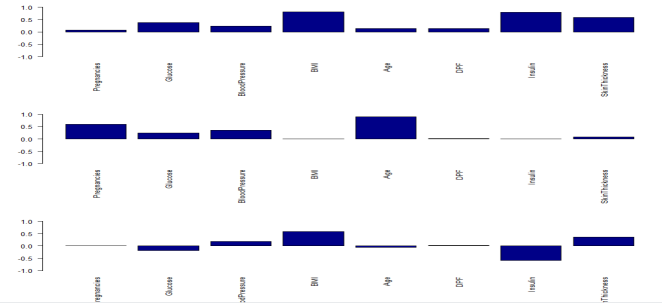




Lastly, as we can see to the left, the first principal component (PC1) appears to be a good differentiator between Individuals with diabetes and those without, as the majority of those that do not suffer the illness fall in the positive spectrum of it, while those who do are on the negative side, capturing that PC1 is a significant, interpretable trend related to having diabetes.

Factor analysis

Factor loadings show how strongly each variable relates to a factor. Larger values mean a stronger connection. Uniqueness values indicate how much of a variable's variance is not explained by the factors; lower uniqueness means the factor explains more of the variable's variation, while higher uniqueness means the variable is mostly independent from the factors.



We performed a factor analysis using both two and three factors to understand the underlying structure of the data. Comparing these solutions provides insight into which model best fits the observed variables.

With two factors, the model explains 39% of the variance. Factor 1 is strongly associated with BMI, SkinThickness, and Insulin (metabolic/body composition features), while Factor 2 mainly represents Age (demographic aspect). However, several variables, such as DPF, Glucose, and Insulin exhibit high uniqueness values, indicating substantial variability left unexplained. The test for model sufficiency yields a very low p-value, suggesting two factors are not enough to capture all patterns in the data.

When using three factors, cumulative variance explained increases to 52%. The core structure remains: Factor 1 (metabolic), Factor 2 (age/demographic), but Factor 3 adds further differentiation, especially for variables like Insulin and SkinThickness. Most uniqueness values decrease, meaning more variance is captured by the model. The chi-square test now gives a non-significant p-value (0.168), showing that a three-factor model adequately fits the data, little structure remains unexplained.

```
Call:
factanal(x = X, factors = 2, scores = "Bartlett", rotation = "varimax")

Uniquenesses:
  Pregnancies       Glucose BloodPressure           BMI           Age           DPF       Insulin
        0.655         0.863         0.792         0.133         0.131         0.982         0.872
SkinThickness
        0.454

Loadings:
              Factor1 Factor2
Pregnancies           0.585
Glucose       0.224   0.294
BloodPressure 0.287   0.355
BMI           0.931
Age                   0.931
DPF           0.132
Insulin       0.329   0.139
SkinThickness 0.730   0.112

                Factor1 Factor2
SS loadings       1.662   1.456
Proportion Var    0.208   0.182
Cumulative Var    0.208   0.390

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 112.99 on 13 degrees of freedom.
The p-value is 4.85e-18
```

```
Call:
factanal(x = X, factors = 3, rotation = "none")

Uniquenesses:
  Pregnancies       Glucose BloodPressure           BMI           Age           DPF       Insulin
        0.642         0.759         0.787         0.005         0.159         0.980         0.005
SkinThickness
        0.516

Loadings:
              Factor1 Factor2 Factor3
Pregnancies           0.593
Glucose       0.387   0.242  -0.184
BloodPressure 0.229   0.359   0.178
BMI           0.807           0.587
Age           0.134   0.905
DPF           0.139
Insulin       0.804          -0.591
SkinThickness 0.590           0.357

                Factor1 Factor2 Factor3
SS loadings       1.890   1.366   0.891
Proportion Var    0.236   0.171   0.111
Cumulative Var    0.236   0.407   0.518

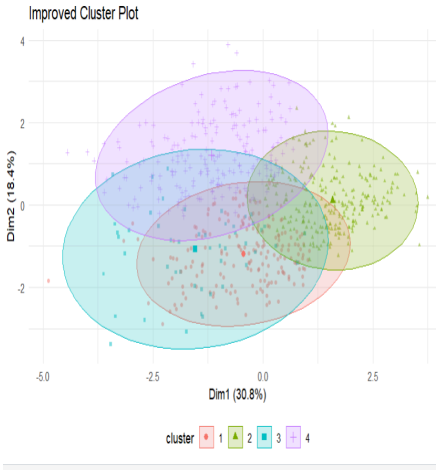
Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 10.39 on 7 degrees of freedom.
The p-value is 0.168
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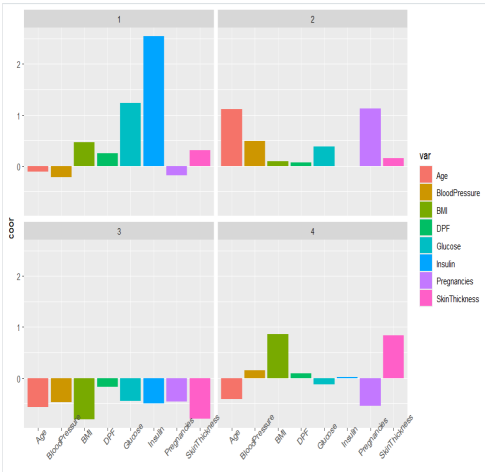
Low overall correlations between most clinical variables in the correlation matrix limit the interpretability and impact of the factor analysis results. Because only a few variable pairs exhibit moderate relationships, the analysis reveals weak underlying latent structure and a relatively small proportion of total variance explained by the factors. This highlights that, while factor analysis can be performed, it may offer only limited clinical insight when applied to data that lack strong inter-variable associations(because the interest variable is categorical)

Clustering tools

Clustering is an unsupervised learning method that finds groups in the data based on similarity. Several techniques were used to determine the most appropriate number of clusters and to evaluate how well the groups are separated.

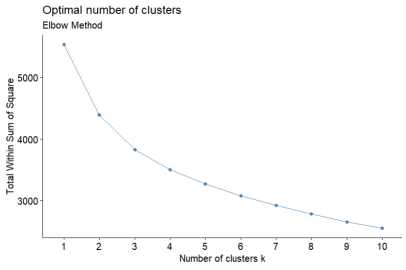
We initially started by setting k=4 for the number of clusters, but these plots reveal that this choice leads to considerable overlap between the clusters. The cluster visualization, barplots of cluster centers, and the low average **silhouette** width (0.18) all indicate that the patient groups are not clearly separated and distinct when using four clusters. This suggests that k=4 is too large for this dataset.

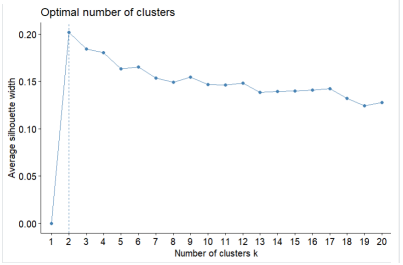


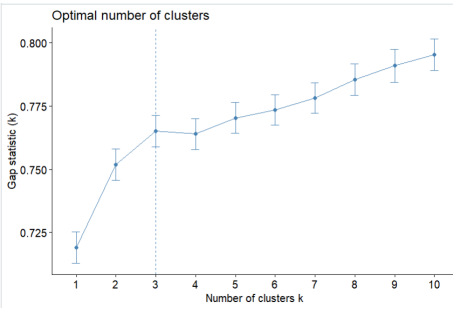




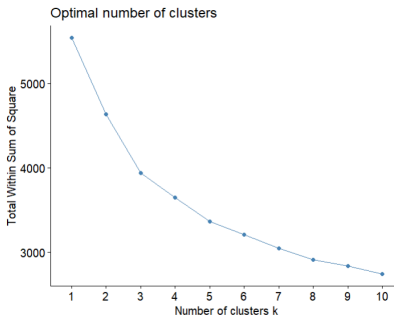
We used several different methods to determine the optimal number of clusters for our dataset. The **elbow method** shows a clear bend at k = 2 or 3. The **silhouette** plot indicates that the quality of clustering is maximized for k = 2. The **gap statistic** also supports k = 2 or k = 3, with a marked improvement at these values before the curve levels off. Overall, all three graphical methods indicate that two or at most three clusters would be optimal for this dataset, providing clear separation and avoiding unnecessary complexity or overlap.

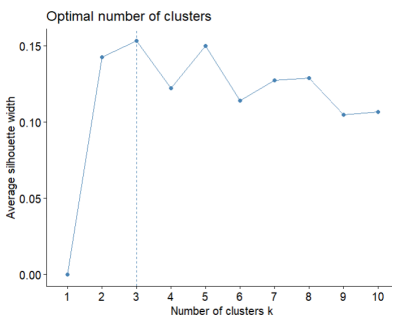


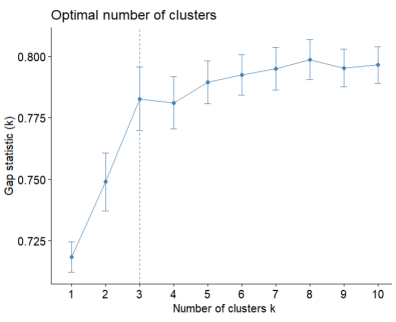




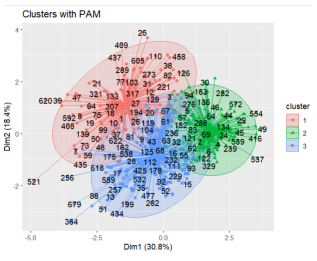
The same for PAM(Partitioning Around Medoids)

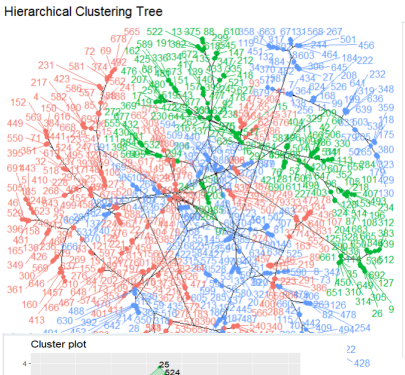


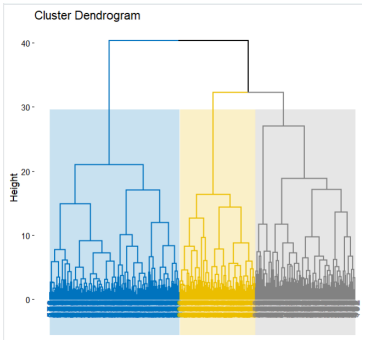




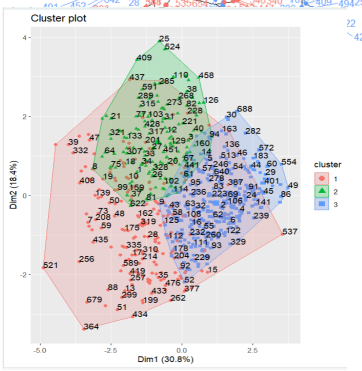
PAM is a clustering method similar to k-means, but instead of using means, each cluster is represented by an actual data point (the medoid). The algorithm assigns points to their nearest medoid, then iteratively swaps medoids with other points if it reduces overall distance, until the best set of medoids and clusters is found. PAM is more robust to outliers than k-means because it uses real observations as centers.
The optimal numbre of clustersPAM is 3



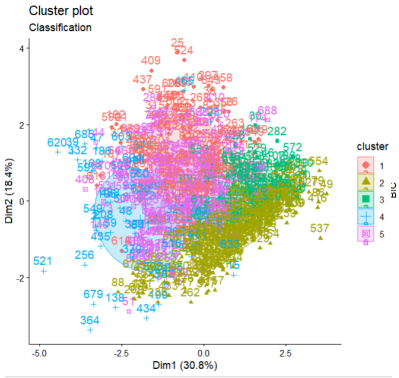


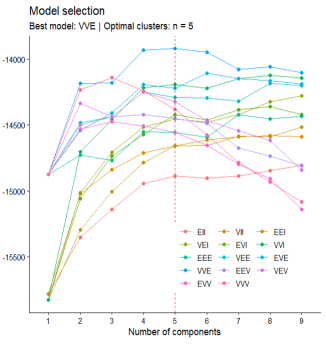


We applied **hierarchical clustering** and visualized the results using a phylogenic tree and a dendrogram. After examining the structure, we found that selecting three clusters (k=3) is optimal, as this solution captures the main divisions in the data without fragmenting groups excessively. The branches for k=3 are well separated, indicating that the clusters are distinct and interpretable, while solutions with more clusters would create artificial splits with little additional benefit.

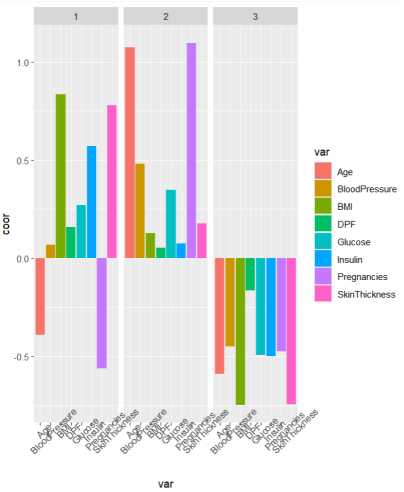


We applied **kernel k-means** clustering, an extension of the classic k-means algorithm that uses a kernel function to allow the detection of clusters that are not linearly separable in the original feature space. With three clusters, the algorithm produced groups of reasonable and similar sizes, indicating a balanced partition of the data. The within-cluster sum of squares values show that each cluster is relatively tight, with no group being overly dispersed. The resulting cluster plot confirms that kernel k-means successfully identifies clear subgroups, even though there remains some overlap between groups, reflecting the complexity of the clinical data. In conclusion, kernel k-means enables the detection of non-linear structures, and in this case, provides a meaningful grouping of the observations that may not be achievable by standard k-means.

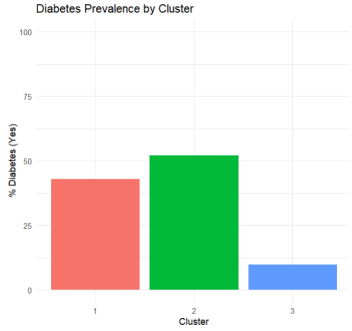




With **EM clustering**, the optimal number of clusters suggested by the Bayesian Information Criterion (BIC) is five. This result differs from k-means, kernel k-means, or hierarchical clustering, which favored two or three clusters. EM clustering is capable of capturing more complex and overlapping group structures due to its probabilistic modeling and flexibility in fitting clusters of varied shapes and sizes. The identification of five clusters means that the EM algorithm finds additional fine-grained subgroups in the data that are not detected by the other methods.



Cluster 1 is characterized by high values of BMI, Glucose, DPF, Insulin, and SkinThickness, but relatively lower Age, suggesting a metabolically altered group with increased risk factors, possibly corresponding to individuals with higher obesity and signs of insulin resistance or poor glycemic control. Cluster 2 stands out for elevated Age and Pregnancies, along with moderately high Blood Pressure, indicating a group of older participants—likely with higher cardiovascular or obstetric risk, but less pronounced metabolic disturbances. Cluster 3 shows uniformly low values across variables, representing the healthiest subgroup in the sample, with lower metabolic and cardiovascular risk indicators. This interpretation provides a clear clinical stratification, distinguishing groups by dominant risk profiles and helping guide further study or interventions.



The barplot summarizes the percentage of patients diagnosed with diabetes ("Yes") within each cluster formed by k-means clustering. Cluster 2 shows the highest diabetes prevalence, with over half of its members affected. Cluster 1 also exhibits a relatively high proportion, around 40%. In contrast, Cluster 3 has a much lower rate of diabetes cases, suggesting it may represent a group with healthier metabolic profiles or lower clinical risk. This differentiation highlights that unsupervised clustering based solely on clinical variables successfully separates subpopulations with distinct diabetes risks, supporting the clinical relevance of the groupings.